Task1_1

```
<class 'PIL.PngImagePlugin.PngImageFile'>
PNG
RGB
(464, 464)
```

I learned how to load images and print the type, format, mode and size of images on google colab.

Task1_2

```
<class 'PIL.PngImagePlugin.PngImageFile'>
float32
(464, 464, 3)
<class 'PIL.PngImagePlugin.PngImageFile'>
```

I learned how to convert an image into a numerical array and back again, showing that computers process images as numbers rather than pictures.

Task1_3

```
<class 'PIL.Image.Image'>
None
RGB
(464, 464)
```

ST1 > Week_6_TutLab

Type | People | Modified | Source

Name	Owner	Date modified	File size
insect1Grayscale.PNG	me	5:39 PM me	87 KB
insect1.PNG	me	Mar 12 me	419 KB

I learned how to convert an image into grayscale, save it as a new file, and reload it to confirm the changes were applied correctly.

Task1_4

```
Downloading data from https://storage.googleapis.com/tensorflow/tf-keras-datasets/mnist.npz
11490434/11490434 ━━━━━━━━━━━━━━━━━━━━ 0s 0us/step
Train (60000, 28, 28) (60000,)
Test ((10000, 28, 28), (10000,))
Train 0 255 33.318421449829934 78.56748998339798
Train 0 255 33.791224489795916 79.17246322228644
```

I learned how to load the MNIST dataset and summarize the values to understand the data before training a model.

Task1_5

```
Train min=0.000, max=255.000
Test min=0.000, max=255.000
Batches train=938, test=157
Batch shape=(64, 28, 28, 1), min=0.000, max=1.000
```

I learned how to reshape the MNIST images, scale their pixel values between 0 and 1, and organize them into batches for training a model.

Task1_6

```
Means train=33.318, test=33.791
Data Generator Mean: 33.318
(64, 28, 28, 1) 0.8190083
(60000, 28, 28, 1) -1.9512918e-05
```

I learned how to adjust the MNIST images so their pixels are centered to help the model learn better.

Task1_7

```
Statistics train=33.318 (78.567), test=33.791 (79.172)
Data Generator mean=33.318, std=78.567
(64, 28, 28, 1) 0.008530649 1.011445
(60000, 28, 28, 1) -3.4560264e-07 0.9999998
```

I learned how to standardize the MNIST images so their pixels have a mean of zero and a standard deviation of one, which helps the model train more effectively